Exploring the Use of Differential Equations in Electric Circuits

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Abstract

We derive and analyze ordinary differential equations (ODEs) governing series RLC circuits to show how resistance controls damping and stability. First-order ODEs governing simplified RC and RL sub-circuits, as well as the second-order ODE describing the full RLC system, are derived and solved analytically for all three configurations, and the role of resistance in controlling circuit damping is examined by varying component parameters. The ideal model is then extended numerically to include current-dependent resistor heating, inductor saturation, and resonant AC forcing. These nonlinear extensions are phenomenological models chosen to illustrate how stress-dependent component behavior can alter transient damping. Our central research question asks: *How do resistors influence the damping and stability of series RLC circuits as described by ODEs, and what are the practical implications for engineering applications such as filters and oscillators?* The analysis identifies resistance as the parameter that governs whether the circuit is overdamped, critically damped, or underdamped, while the numerical results show how non-ideal components can suppress, shift, or amplify transient oscillations.

Index Terms

series RLC circuit, ordinary differential equations, damping, Kirchhoff's Voltage Law, stability analysis, nonlinear simulation, signal processing.

Nomenclature

R	Series resistance, measured in Ω.
L	Series inductance, measured in H.
C	Series capacitance, measured in F.
$V(t)$	Applied source voltage.
V_s	Step-source amplitude.
$v_C(t)$	Capacitor voltage, the primary output variable.
$\dot{v}_C(t)$	Time derivative of capacitor voltage, dv_C/dt, measured in V/s.
$i(t)$	Series current through the resistor, inductor, and capacitor.
α	Damping coefficient, $\alpha = R/(2L)$.
ω_0	Undamped natural frequency, $\omega_0 = 1/\sqrt{LC}$.
ω_d	Damped natural frequency for the underdamped case.
R_{crit}	Critical resistance, $R_{\text{crit}} = 2\sqrt{L/C}$.
R_0	Nominal resistance used in the non-ideal simulation.
L_0	Nominal inductance used in the non-ideal simulation.
k	Heating coefficient in the current-dependent resistance model, measured in Ω/A^2.
I_{sat}	Saturation current threshold for the inductor model.
y_1, y_2	State variables used in the numerical simulation.

I. Introduction

The dynamic behavior of such circuits is naturally captured by *ordinary differential equations* (ODEs). Simpler RC and RL sub-circuits obey *first-order* ODEs, while the complete RLC system requires a *second-order* ODE to describe the interplay between energy stored in the inductor's magnetic field and

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TABLE I
NOMINAL COMPONENT VALUES

Quantity	Symbol	Nominal value
Source amplitude	V_s	10 V
Resistance cases	R	50 Ω, 200 Ω, 500 Ω
Inductance	L	100 mH = 0.1 H
Capacitance	C	$10\,\mu$F = 10×10^{-6} F

the capacitor's electric field. Understanding the transition from first- to second-order models clarifies why the resistor plays an indispensable role: it is the only purely dissipative element in the circuit.

Our scientific question is:

How do resistors influence the damping and stability of series RLC circuits as described by ODEs, and what are the practical implications in engineering applications?

This paper answers that question through a two-part approach: first by deriving ideal analytic solutions for RC, RL, and RLC circuits, and second by comparing those ideal responses with numerical simulations that include phenomenological non-ideal resistor and inductor behavior.

The remainder of this paper is structured as follows. Section II introduces the circuit configuration and the governing physical principles. Section III details the derivation of the governing equations, identification of variables and parameters, and the analytic solutions for each circuit type. Section IV presents results from varying circuit parameters and initial conditions. Section V extends the ideal model to non-ideal components — specifically current-dependent resistor heating and inductor saturation — and resonant AC forcing, using numerical simulation. Section VI interprets the findings in the context of our research question, discusses limitations, and proposes future work. Section VII concludes the paper.

II. System Model and Governing Principles

A. Circuit Overview

We consider a series RLC circuit driven by a step voltage source $V(t)$. At $t = 0$, a DC source of amplitude V_s is connected to an initially uncharged capacitor, so $V(t) = V_s$ for the post-switching interval $t \geq 0$. The quantity of primary interest is the capacitor voltage $v_C(t)$ for $t \geq 0$. The component values used for the main RLC case study are summarized in Table I.

These values were chosen because they produce simple, transparent damping parameters while remaining realistic for a bench-top laboratory circuit: $LC = 1 \times 10^{-6}\,\mathrm{s}^2$, so $\omega_0 = 1/\sqrt{LC} = 1000\,\mathrm{rad/s}$, and $R_{\mathrm{crit}} = 2\sqrt{L/C} = 200\,\Omega$. Thus $R = 50\,\Omega$, $R = 200\,\Omega$, and $R = 500\,\Omega$ provide clear underdamped, critically damped, and overdamped examples, respectively.

B. State Variables

The two state variables are the capacitor voltage $v_C(t)$ and the inductor current $i(t)$, which also equals the series current through every element. These quantities are chosen because neither can change instantaneously: the capacitor voltage is proportional to stored charge ($v_C = q/C$), and the inductor current is proportional to stored magnetic flux ($\Psi = Li$). Consequently, they fully characterize the energy state of the circuit at any instant.

C. Initial Conditions

Before the switch is closed at $t = 0$:

- The capacitor is initially uncharged: $v_C(0) = 0\,\mathrm{V}$.
- No current flows: $i(0) = 0\,\mathrm{A}$.

Prior to switching, the capacitor behaves as an open circuit (blocking DC current) and the ideal inductor behaves as a short circuit (zero DC voltage drop).

D. Governing Equations

1) Kirchhoff's Voltage Law: Applying Kirchhoff's Voltage Law (KVL) around the series loop [1]:

$$Ri + L\frac{di}{dt} + v_C = V(t). \tag{1}$$

2) Second-Order ODE for the RLC Circuit: Using the capacitor voltage relationship $i(t) = C\,dv_C/dt$ and substituting into (1) yields the governing second-order ODE:

$$LC\frac{d^2v_C}{dt^2} + RC\frac{dv_C}{dt} + v_C = V(t). \tag{2}$$

3) First-Order ODEs for Reduced Circuits: For an RC circuit (inductor replaced by an ideal wire, $L \to 0$):

$$\frac{dv_C}{dt} + \frac{1}{RC}v_C = \frac{V_s}{RC}. \tag{3}$$

For an RL circuit (capacitor replaced by an ideal wire):

$$\frac{di}{dt} + \frac{R}{L}i = \frac{V_s}{L}. \tag{4}$$

III. Analytical Methods

A. Equations, Conditions, and Connection to the Application

The three ODEs derived above–(3), (4), and (2)–together model the spectrum from simple to complex circuit behavior. With initial conditions applied, the initial value problems (IVPs) are:

RC circuit (first-order):

$$\frac{dv_C}{dt} + \frac{1}{RC}v_C = \frac{V_s}{RC}, \qquad v_C(0) = V_0. \tag{5}$$

RL circuit (first-order):

$$\frac{di}{dt} + \frac{R}{L}i = \frac{V_s}{L}, \qquad i(0) = 0. \tag{6}$$

RLC circuit (second-order):

$$LC\frac{d^2v_C}{dt^2} + RC\frac{dv_C}{dt} + v_C = V_s, \qquad v_C(0) = 0\,\mathrm{V}, \quad \dot{v_C}(0) = 0. \tag{7}$$

Each equation models a physical balance: the left-hand terms represent energy dissipation (via R) and energy storage (via L and C), while the right-hand side represents the forcing from the voltage source.

B. Independent and Dependent Variables

- **Independent variable:** time t in s.
- **Dependent variables:** $v_C(t)$ in V (capacitor voltage) and $i(t)$ in A (series current).

C. Parameters

- R: resistance; dissipates electrical energy as heat. Appears in the first-order damping term R/L and the second-order damping coefficient $\alpha = R/(2L)$.
- $L = 0.1\,\mathrm{H}$: inductance; stores energy in a magnetic field. Governs how quickly current can change.
- $C = 10 \times 10^{-6}\,\mathrm{F}$: capacitance; stores energy in an electric field. Governs how quickly voltage can change.
- $V_s = 10\,\mathrm{V}$: step-input amplitude. Represents a DC voltage switched on at $t = 0$.

D. Assumptions and Limitations

a) Assumptions: We adopt the standard *lumped-element model*: all resistance is concentrated in R, all inductance in L, and all capacitance in C, with no parasitic coupling between components. Component values are assumed to be constant with respect to temperature, frequency, and signal amplitude (i.e., linear, time-invariant behavior). Wires and switch contacts are treated as ideal (zero resistance, zero inductance).

b) Limitations: Real resistors exhibit temperature-dependent resistance governed by $R(T) = R_0[1 + \alpha_R(T - T_0)]$, introducing nonlinear behavior under large current loads. At high frequencies, parasitic inductance in capacitors and parasitic capacitance in inductors violate the lumped-element assumption, requiring distributed-element or transmission-line models. The ideal step-input assumption ignores source impedance and switching transients present in physical systems. Section V relaxes two of these assumptions by simulating current-dependent resistor heating and inductor saturation.

E. Analytic Solutions

1) First-Order RC Circuit: Equation (5) is a linear first-order ODE with constant coefficients. Using the integrating factor $\mu(t) = e^{t/(RC)}$, the general solution is:

$$v_C(t) = V_s + (V_0 - V_s)\, e^{-t/(RC)}. \tag{8}$$

The time constant $\tau_{RC} = RC$ determines how quickly v_C relaxes to the steady-state value V_s.

2) First-Order RL Circuit: Similarly, (6) yields:

$$i(t) = \frac{V_s}{R}\left(1 - e^{-(R/L)t}\right). \tag{9}$$

The time constant is $\tau_{RL} = L/R$.

3) Second-Order RLC Circuit: Characteristic Equation and Damping: Seeking homogeneous solutions of the form e^{rt} in (7) leads to the characteristic equation:

$$LCr^2 + RCr + 1 = 0. \tag{10}$$

Dividing by LC and introducing the standard parameters

$$\alpha = \frac{R}{2L}, \qquad \omega_0 = \frac{1}{\sqrt{LC}}, \tag{11}$$

the roots become $r = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$. The discriminant $\Delta = \alpha^2 - \omega_0^2$ determines damping behavior:

- **Overdamped** ($\Delta > 0$, i.e. $R > 2\sqrt{L/C}$): two distinct real negative roots. The circuit returns to equilibrium without oscillating, but slowly. This occurs in high-resistance configurations such as lossy filters.
- **Critically damped** ($\Delta = 0$, i.e. $R = R_{\text{crit}} = 2\sqrt{L/C}$): one repeated real root $r = -\alpha$. Fastest non-oscillatory return to equilibrium; used in precision instruments that must settle quickly.
- **Underdamped** ($\Delta < 0$, i.e. $R < 2\sqrt{L/C}$): two complex conjugate roots $r = -\alpha \pm j\omega_d$, where $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$ is the damped natural frequency. The circuit oscillates with exponentially decaying amplitude; typical in resonant and tuned circuits.

4) Second-Order RLC Circuit: General Analytic Solution:

a) Homogeneous solutions: For each damping case, the general homogeneous solution is:

Overdamped (r_1, r_2 real, distinct):

$$v_{C,h}(t) = A_1 e^{r_1 t} + A_2 e^{r_2 t}. \tag{12}$$

Critically damped ($r = -\alpha$ repeated):

$$v_{C,h}(t) = (A_1 + A_2 t)\, e^{-\alpha t}. \tag{13}$$

Underdamped (complex roots):

$$v_{C,h}(t) = e^{-\alpha t}(A_1 \cos \omega_d t + A_2 \sin \omega_d t)\,. \tag{14}$$

b) Non-homogeneous solution: For a constant forcing $V(t) = V_s$, the particular solution is simply $v_{C,p} = V_s$. For time-varying inputs, two systematic methods apply: (i) *Variation of parameters*, which replaces the constants A_1, A_2 with functions $u_1(t), u_2(t)$ determined by a two-equation system derived from the ODE. For example, in the underdamped case one may write

$$v_C(t) = V_s + u_1(t)e^{-\alpha t}\cos\omega_d t + u_2(t)e^{-\alpha t}\sin\omega_d t, \tag{15}$$

where $u_1(t)$ and $u_2(t)$ are chosen so that the forced equation and its auxiliary constraint are both satisfied. This converts a nonconstant forcing term into first-order equations for u_1 and u_2 [1]. Alternatively, (ii) *Laplace transforms*, which convert the ODE to an algebraic equation $\mathcal{L}\{v_C\}(s) = \mathcal{L}\{V(t)\}(s) \cdot H(s)$, where $H(s) = 1/(LCs^2 + RCs + 1)$ is the circuit transfer function, followed by a partial-fraction expansion and inverse transform [2].

c) Applying initial conditions (underdamped case): Here V_0 denotes the general initial capacitor voltage; the main case study in Table I uses $V_0 = 0$, consistent with the zero-charge initial condition in (7). The complete solution is $v_C(t) = v_{C,h}(t) + V_s$. Applying $v_C(0) = V_0$ and $\dot{v_C}(0) = 0$:

$$v_C(0) = A_1 + V_s = V_0 \implies A_1 = V_0 - V_s, \tag{16}$$

$$\dot{v_C}(0) = -\alpha A_1 + \omega_d A_2 = 0 \implies A_2 = \frac{\alpha(V_0 - V_s)}{\omega_d}. \tag{17}$$

Substituting (16)–(17) into (14) and adding V_s gives the complete analytic underdamped step response:

$$v_C(t) = V_s + (V_0 - V_s)e^{-\alpha t}\left(\cos\omega_d t + \frac{\alpha}{\omega_d}\sin\omega_d t\right). \tag{18}$$

For the typical step-response case $V_0 = 0$, this simplifies to $v_C(t) = V_s\left[1 - e^{-\alpha t}\left(\cos\omega_d t + \frac{\alpha}{\omega_d}\sin\omega_d t\right)\right]$.

For completeness, the initial conditions $v_C(0) = 0$ and $\dot{v_C}(0) = 0$ are applied to the remaining two cases. For the critically damped case, the complete solution is $v_C(t) = v_{C,h}(t) + V_s$ with $v_{C,h}(t) = (A_1 + A_2 t)\, e^{-\alpha t}$. Applying the initial conditions:

$$v_C(0) = A_1 + V_s = 0 \implies A_1 = -V_s, \tag{19}$$

$$\dot{v_C}(0) = -\alpha A_1 + A_2 = 0 \implies A_2 = -\alpha V_s, \tag{20}$$

giving the complete critically damped step response:

$$v_C(t) = V_s\left[1 - (1 + \alpha t)\, e^{-\alpha t}\right]. \tag{21}$$

For the overdamped case, $v_{C,h}(t) = A_1 e^{r_1 t} + A_2 e^{r_2 t}$. Applying the initial conditions:

$$v_C(0) = A_1 + A_2 + V_s = 0 \implies A_1 + A_2 = -V_s, \tag{22}$$

$$\dot{v_C}(0) = r_1 A_1 + r_2 A_2 = 0, \tag{23}$$

and solving this 2×2 system:

$$A_1 = -V_s\,\frac{r_2}{r_2 - r_1}, \qquad A_2 = V_s\,\frac{r_1}{r_2 - r_1}, \tag{24}$$

giving the complete overdamped step response:

$$v_C(t) = V_s\left[1 - \frac{r_2\, e^{r_1 t} - r_1\, e^{r_2 t}}{r_2 - r_1}\right]. \tag{25}$$

IV. Results

A. First-Order RC Response

For the RC sub-circuit, the governing equation can be written as

$$\dot{v_C} = \frac{V_s - v_C}{RC}. \tag{26}$$

The equilibrium is $v_C^* = V_s = 10\,\mathrm{V}$. If $v_C < V_s$, then $\dot{v_C} > 0$; if $v_C > V_s$, then $\dot{v_C} < 0$. Thus the phase line points toward v_C^* from both sides, so the equilibrium is asymptotically stable.

Using $R = 100\,\Omega$ and $C = 10\,\mu\mathrm{F}$ (a separate RC sub-circuit example, independent of the RLC values in Table I) gives $\tau_{RC} = RC = 1\,\mathrm{ms}$. For three initial capacitor voltages,

$$v_C(t; V_0 = 0) = 10\left(1 - e^{-1000t}\right), \tag{27}$$

$$v_C(t; V_0 = 5) = 10 - 5e^{-1000t}, \tag{28}$$

$$v_C(t; V_0 = 10) = 10. \tag{29}$$

All three cases approach the same stable equilibrium, but they start from different points on the phase line.

With $V_0 = 0$ fixed and $C = 10\,\mu\mathrm{F}$, varying resistance changes only the time constant:

$$R = 50\,\Omega: \quad v_C(t) = 10\left(1 - e^{-2000t}\right), \tag{30}$$

$$R = 100\,\Omega: \quad v_C(t) = 10\left(1 - e^{-1000t}\right), \tag{31}$$

$$R = 200\,\Omega: \quad v_C(t) = 10\left(1 - e^{-500t}\right). \tag{32}$$

Larger resistance produces slower charging, which is the first-order analog of the damping effect studied in the full RLC circuit.

B. First-Order RL Response

For the RL sub-circuit, the analytic solution (9) can similarly be evaluated for three values of resistance with $L = 0.1\,\mathrm{H}$ and $V_s = 10\,\mathrm{V}$:

$$R = 50\,\Omega: \quad i(t) = 0.2\left(1 - e^{-500t}\right)\ \mathrm{A}, \tag{33}$$

$$R = 100\,\Omega: \quad i(t) = 0.1\left(1 - e^{-1000t}\right)\ \mathrm{A}, \tag{34}$$

$$R = 200\,\Omega: \quad i(t) = 0.05\left(1 - e^{-2000t}\right)\ \mathrm{A}. \tag{35}$$

Larger resistance both reduces the steady-state current V_s/R and increases the rate of rise (since $\tau_{RL} = L/R$ decreases), the inverse of the RC behavior where larger R slows the response.

C. Second-Order RLC Response

For the selected component values,

$$\omega_0 = 1000\,\mathrm{rad/s}, \qquad R_{\mathrm{crit}} = 200\,\Omega. \tag{36}$$

The three resistance values in Table II therefore sample the underdamped, critically damped, and overdamped regimes without changing L, C, or the source voltage.

With $v_C(0) = 0$, $\dot{v_C}(0) = 0$, and $V_s = 10\,\mathrm{V}$, the numerical step responses are as follows, with t measured in seconds and v_C measured in volts:

$$v_{C,50}(t) = 10\left[1 - e^{-250t}(\cos(968.25t) + 0.2582\sin(968.25t))\right], \tag{37}$$

$$v_{C,200}(t) = 10\left[1 - (1 + 1000t)e^{-1000t}\right], \tag{38}$$

$$v_{C,500}(t) = 10 - 10.456e^{-208.71t} + 0.456e^{-4791.29t}. \tag{39}$$

TABLE II
DAMPING CASES FOR THE SERIES RLC STEP RESPONSE

Case	R	α	Characteristic data
Underdamped	$50\,\Omega$	$250\,\mathrm{s}^{-1}$	$\omega_d = 968.25\,\mathrm{rad/s}$
Critically damped	$200\,\Omega$	$1000\,\mathrm{s}^{-1}$	$r = -1000\,\mathrm{s}^{-1}$
Overdamped	$500\,\Omega$	$2500\,\mathrm{s}^{-1}$	$r_1 = -208.71\,\mathrm{s}^{-1}$, $r_2 = -4791.29\,\mathrm{s}^{-1}$

Equation (37) oscillates around the final value before settling, (38) reaches the final value as quickly as possible without oscillation, and (39) approaches the final value without overshoot but is slowed by the dominant pole $r_1 = -208.71\,\mathrm{s}^{-1}$.

The phase trajectory in Fig. 2 uses the state coordinates $(v_C, \dot{v_C})$. In these coordinates the step response equilibrium is $(V_s, 0) = (10\,\mathrm{V}, 0\,\mathrm{V/s})$, not the origin. Equivalently,

$$\dot{x}_1 = x_2, \qquad \dot{x}_2 = \frac{V_s - x_1 - RCx_2}{LC}, \quad x_1 = v_C,\ x_2 = \dot{v_C}. \tag{40}$$

For the underdamped case $R = 50\,\Omega$, the eigenvalues are $-250 \pm j\,968.25$, so the trajectory is a stable spiral approaching $(10, 0)$ [1]. If the shifted coordinate $u = v_C - V_s$ were used instead, the same spiral would approach $(u, \dot{u}) = (0, 0)$.

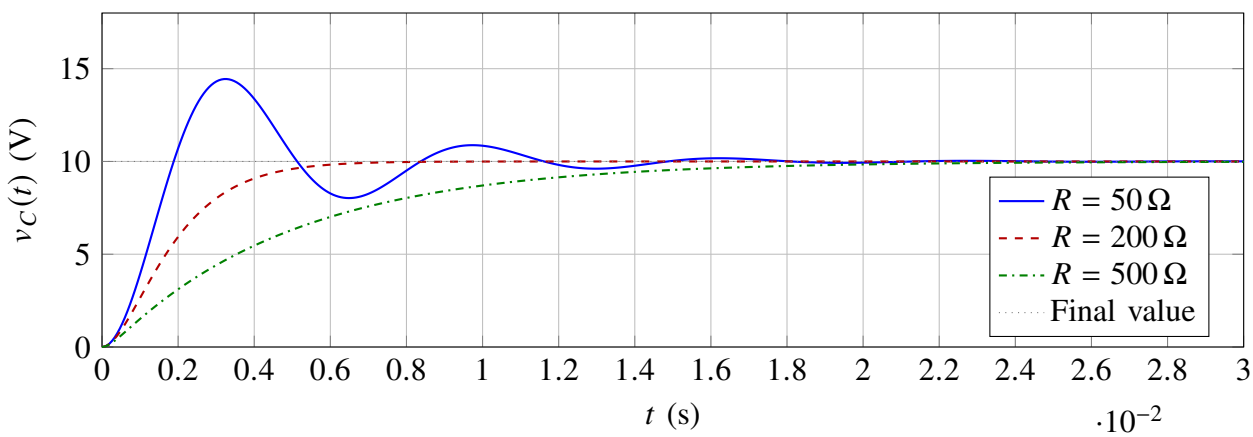

Fig. 1. Capacitor voltage $v_C(t)$ for three values of resistance R, showing underdamped, critically damped, and overdamped responses.

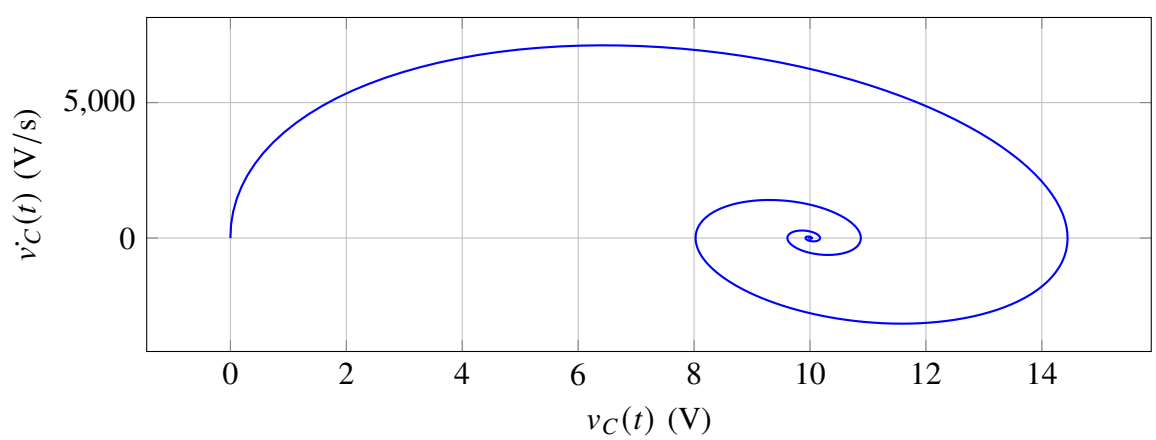

Fig. 2. Phase trajectory of the second-order system ($\dot{v_C}$ vs. v_C) for the underdamped case with $R = 50\,\Omega$, converging to the equilibrium $(v_C, \dot{v_C}) = (10\,\mathrm{V},\ 0\,\mathrm{V/s})$.

D. Connection to the Research Question

The results show that changing R changes the characteristic roots while leaving the energy-storage parameters L and C fixed. For $R = 50\,\Omega$, the roots are complex and the capacitor voltage rings before settling. For $R = 200\,\Omega$, the repeated real root gives the fastest non-oscillatory response. For $R = 500\,\Omega$, two negative real roots remove oscillation but slow the return to equilibrium. Thus resistance directly controls the damping mechanism described in the research question. This ideal-response baseline is the reference point for the non-ideal simulations in Section V.

V. Numerical Simulation of Non-Ideal Components

A. Nonlinear Modeling and State-Space Representation

The analytic solutions derived in the previous sections rely on ideal, constant component values. In practical circuits, however, passive components can change behavior under high-stress transient conditions. To test how these material limits alter the ideal mathematical model, this section adds two phenomenological nonlinear effects: current-dependent heating in the resistor and magnetic saturation in the inductor.

Joule heating causes a resistor's temperature, and therefore its effective resistance, to increase as current flows through it. We model this dynamic resistance as

$$R(i) = R_0 + ki^2, \tag{41}$$

where R_0 is the nominal resistance and k is a heating coefficient. Dimensionally, $[k][i^2] = \Omega$ requires $[k] = \Omega/\mathrm{A}^2$; that is, k carries units of Ω/A^2. Similarly, an inductor with a saturating magnetic core loses effective inductance at high current. This effect is approximated by

$$L(i) = \frac{L_0}{1 + (i/I_{\mathrm{sat}})^2}, \tag{42}$$

where L_0 is the nominal inductance and I_{sat} is the saturation current threshold [3].

Substituting these dynamic parameters into Kirchhoff's Voltage Law gives the effective nonlinear model

$$L(i)C\frac{d^2v_C}{dt^2} + R(i)C\frac{dv_C}{dt} + v_C = V(t), \tag{43}$$

with series current $i(t) = C\,dv_C/dt$. Because the resistive term now contains a cubic dependence on dv_C/dt and the inductive term changes with current, the characteristic equation method used for constant-coefficient systems no longer applies. The functional forms chosen for $R(i)$ and $L(i)$ are phenomenological: they illustrate how state-dependent component behavior can affect the transient response, without claiming to represent a specific commercial device in detail.

For numerical simulation, let $y_1 = v_C$ and $y_2 = dv_C/dt$. Then $i = Cy_2$, and (43) becomes the first-order state-space system

$$\frac{dy_1}{dt} = y_2, \tag{44}$$

$$\frac{dy_2}{dt} = \frac{V(t) - y_1 - R(Cy_2)Cy_2}{L(Cy_2)C}. \tag{45}$$

The system was integrated in Python using an adaptive RK45 method [4] with $R_0 = 50\,\Omega$, $L_0 = 0.1\,\mathrm{H}$, $C = 10\,\mu\mathrm{F}$, $k = 8000\,\Omega/\mathrm{A}^2$, and $I_{\mathrm{sat}} = 0.05\,\mathrm{A}$. These values are not meant to identify a specific commercial component; instead, they isolate how stress-dependent component behavior changes the same baseline RLC circuit studied analytically above.

B. Simulation Results and Physical Interpretation

The numerical simulations in Fig. 3 show that the ideal step response remains a useful baseline, but it does not capture all of the transient behavior produced by stressed real components. Three non-ideal scenarios are examined: thermal resistance, inductor saturation, and resonant AC forcing.

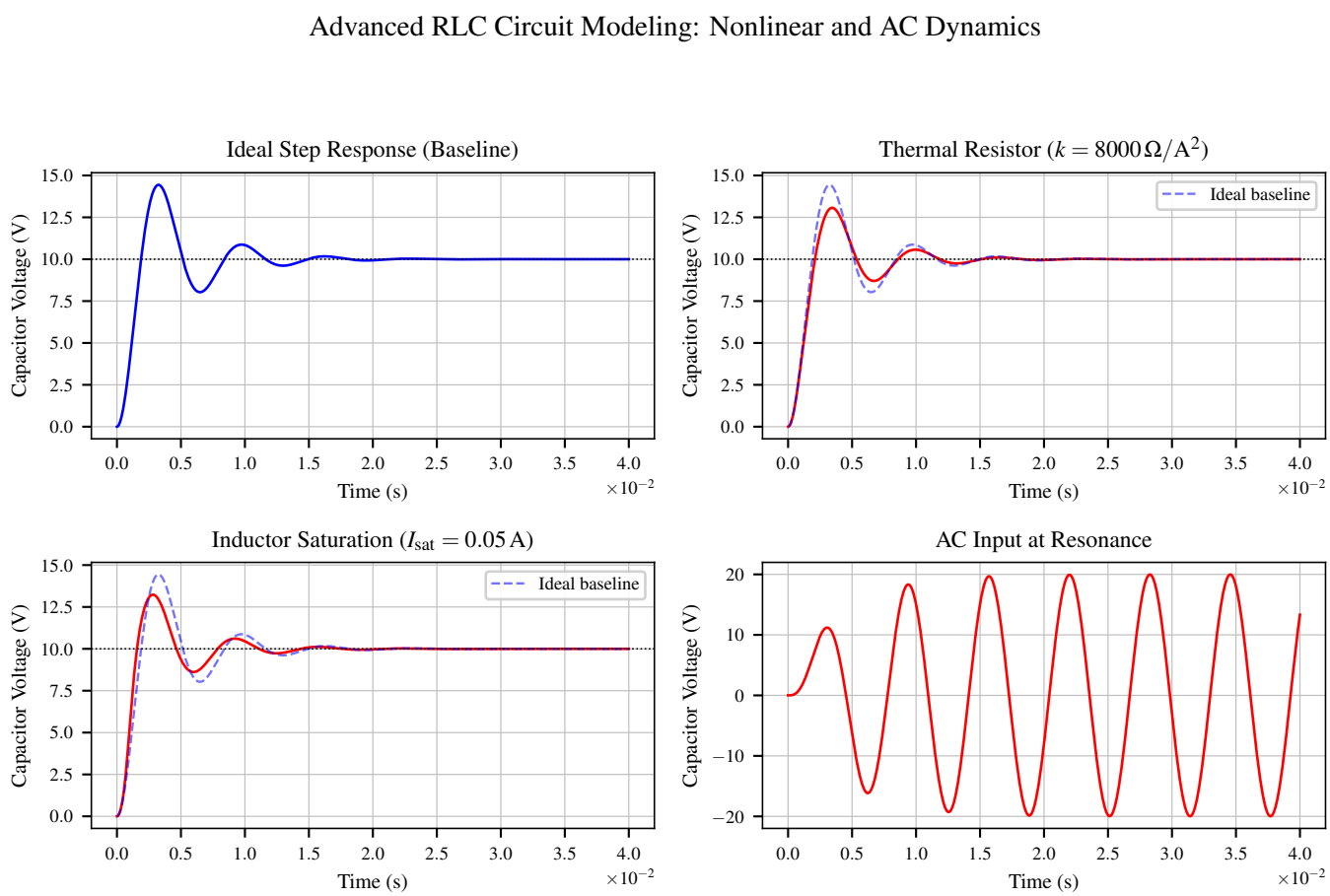

Fig. 3. Numerical step responses of the RLC circuit under non-ideal material constraints compared against the ideal baseline model ($R_0 = 50\,\Omega$, $L_0 = 0.1\,\mathrm{H}$, $C = 10\,\mu\mathrm{F}$).

1) Thermal Resistor Response: During the initial transient interval, the initially uncharged capacitor draws a large inrush current. In the thermal model, this current surge raises the effective resistance above the nominal $50\,\Omega$ value exactly when the circuit has the most stored transient energy. The resistor therefore acts like a dynamic brake: it dissipates energy more aggressively during the first swing, suppressing the overshoot from about 14.44 V in the ideal model to about 13.07 V and bringing the response back toward the 10 V steady state more quickly.

2) Inductor Saturation Response: When inductor saturation is introduced, the response also deviates from the ideal baseline, but for a different reason. The high initial current lowers the effective inductance in (42); since $\omega_0 = 1/\sqrt{LC}$, the instantaneous natural frequency increases while the inductor is saturated. The voltage peak therefore occurs earlier than in the ideal model, and the first overshoot is reduced to about 13.24 V. As the transient current decays, $L(i)$ returns toward L_0, so the response gradually realigns with the ideal curve.

3) AC Input at Resonance: To demonstrate energy buildup, the step input was replaced with an AC source

$$V(t) = V_s \sin(\omega_0 t), \tag{46}$$

driven at the ideal natural frequency $\omega_0 = 1000\,\mathrm{rad/s}$. Because the driving frequency matches the circuit's natural energy exchange between L and C, the capacitor voltage grows far beyond the source amplitude before settling into a large oscillation. In this simulation, the capacitor voltage approaches nearly $\pm 20\,\mathrm{V}$ even though the source amplitude is only 10 V.

These simulations refine the research question. An ideal resistor determines the broad damping regime, but real-world component nonlinearities make damping state-dependent: the resistor, inductor, and input waveform can all change the transient path taken toward stability.

VI. Discussion

A. Summary of Problem and Results

This paper modeled series RLC circuits using first- and second-order ODEs, deriving analytic solutions for RC, RL, and RLC configurations. The resistor was identified as the sole energy-dissipating element, and its value shown to directly determine the discriminant of the characteristic equation (10), thereby controlling whether the circuit is overdamped, critically damped, or underdamped. The numerical extension then showed that stress-dependent components can change the transient path even when the nominal values remain fixed.

B. Interpretation in Terms of the Research Question

Without resistance ($R = 0$), the characteristic roots are purely imaginary ($r = \pm j\omega_0$), yielding perpetual, undamped oscillations in an ideal LC tank circuit. Any nonzero R introduces the decay envelope $e^{-\alpha t}$ that eventually damps oscillations to zero. In engineering applications: bandpass filters rely on controlled underdamping [5] to achieve frequency selectivity when the output is taken across the resistor, while the capacitor-voltage response studied here behaves like a low-pass response with possible resonant overshoot. Overdamped circuits ensure no oscillatory transients (ringing) in power supplies and ADC front-ends [5]; critically damped designs appear in servo systems where settling speed is paramount. A passive RLC circuit can store and exchange energy, but a sustained oscillator also requires an active energy source or feedback element to replace resistive losses. This analytical framework directly addresses the research question: resistors are essential for stabilizing oscillatory behavior and enabling practical circuit design. The non-ideal simulations add an important refinement: damping is not always a constant design parameter during a transient. When current is large, resistor heating can increase dissipation and suppress overshoot, while inductor saturation can temporarily raise the natural frequency and shift the timing of voltage peaks. The resonant AC case also shows that even with resistance present, a matched driving frequency can produce capacitor voltages much larger than the source amplitude.

C. Limitations and Future Work

The lumped-element and linearity assumptions restrict the model's validity to low frequencies and small signals. Future work should incorporate: (i) calibration of k and I_{sat} against measured component data; (ii) a temperature state equation for resistor self-heating rather than the instantaneous approximation $R(i) = R_0 + ki^2$; (iii) distributed-element analysis for RF applications above 1 MHz; (iv) experimental validation using an oscilloscope to compare measured $v_C(t)$ waveforms against the analytic solutions derived here; and (v) extension to parallel RLC and more complex multi-mesh circuits modeled by systems of ODEs.

VII. Conclusion

We derived and solved first- and second-order ODEs governing RC, RL, and series RLC circuits, demonstrating analytically and qualitatively how the resistor controls damping. The three damping regimes– overdamped, critically damped, and underdamped– correspond directly to engineering trade-offs between settling speed and oscillatory behavior. The numerical non-ideal simulations further show that real component behavior can change the shape and timing of the transient response: thermal resistance suppresses overshoot, inductor saturation shifts oscillation timing, and resonant forcing can amplify capacitor voltage beyond the source amplitude. The analysis shows that resistors are not merely optional loss elements but are indispensable for stable, predictable circuit operation, especially when component limits are included in the model.

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